

AI Hype vs Stock asset's price: NVDA (2023–2025)

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Background



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started in 1993 as a graphics chip startup by Jensen Huang, Chris Malachowsky, and Curtis Priem

It produces Graphics Processing Units (GPUs), Systems on Chips (SoCs), and software platforms that power AI, gaming, professional visualization, data centers, and automotive applications, essentially providing the core accelerated computing hardware and platforms for complex digital tasks, especially in Artificial Intelligence. Their main products include GeForce GPUs for gamers, DGX/HGX systems for AI supercomputing, and platforms like Omniverse for industrial digitalization.

Timeline of the company development



1993

3D Graphics

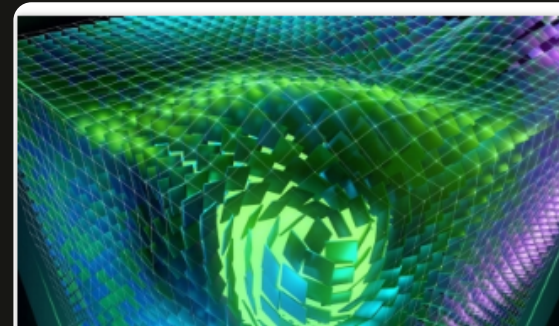
Founded on April 5, 1993, by Jensen Huang, Chris Malachowsky, and Curtis Priem, with a vision to bring 3D graphics to the gaming and multimedia markets.



1999

GPU

Invents the GPU, the graphics processing unit, which sets the stage to reshape the computing industry.



2006

CUDA

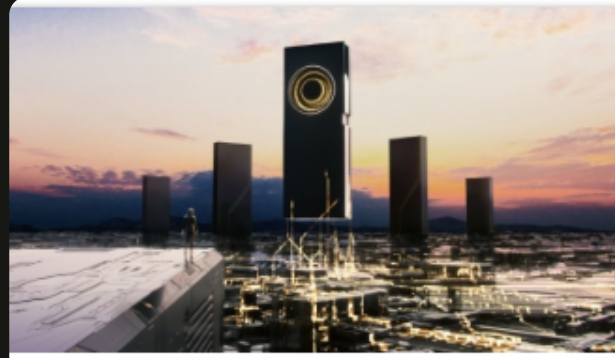
Opens parallel processing capabilities of GPUs to science and research with unveiling of CUDA® architecture.



2012

AI

Sparks the era of modern AI by powering the breakthrough AlexNet neural network.



2018

RTX

Reinvents computer graphics with NVIDIA RTX™, the first GPU capable of real-time ray tracing.

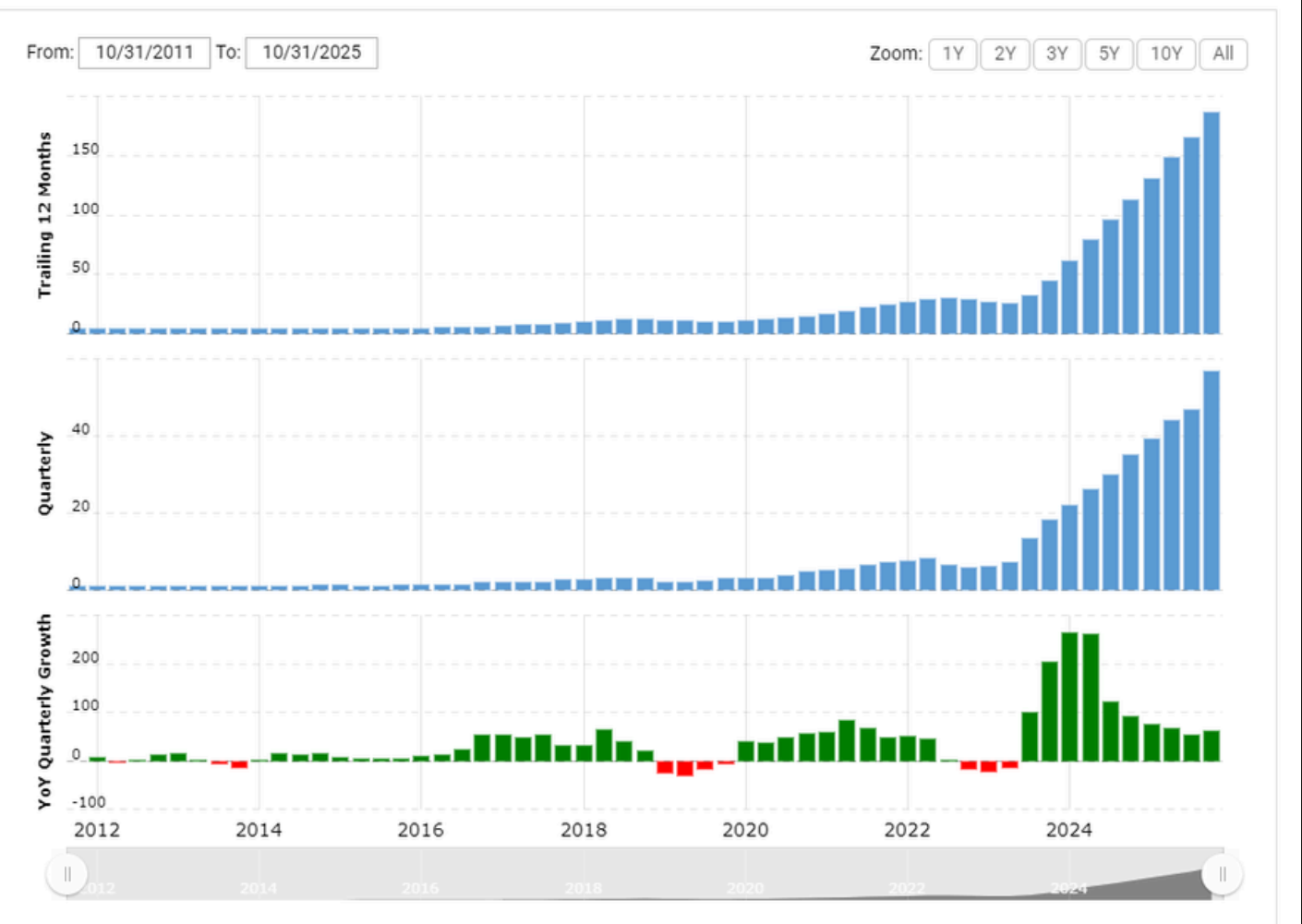


2022

Omniverse

Plays a foundational role in the building of the metaverse, the next stage of the internet, with the NVIDIA Omniverse™ platform.

Market information



Revenue and profit

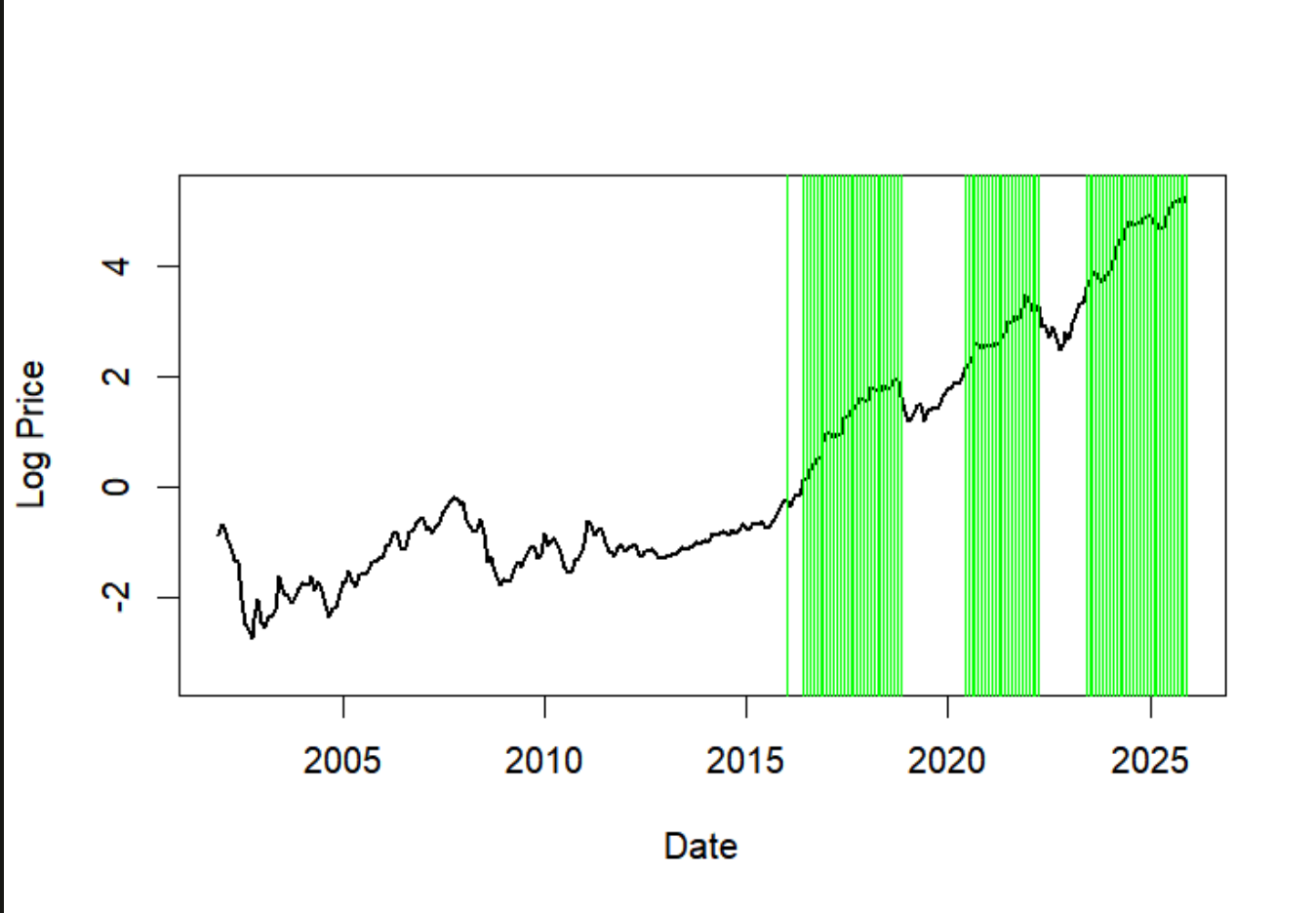
Source: macrotrends

Market Position — Nvidia & Competitors

Discrete GPU Market: Nvidia ~94%, AMD ~6%, Intel ~<1% (Jon Peddie).

AI Chip Market: Nvidia leads with ~80% share; AMD, Intel, Broadcom and TPUs share the remaining ~20%.

Key competitors include AMD (Instinct MI series), Intel (Arc & Gaudi), Google TPU, and Chinese AI chip makers.

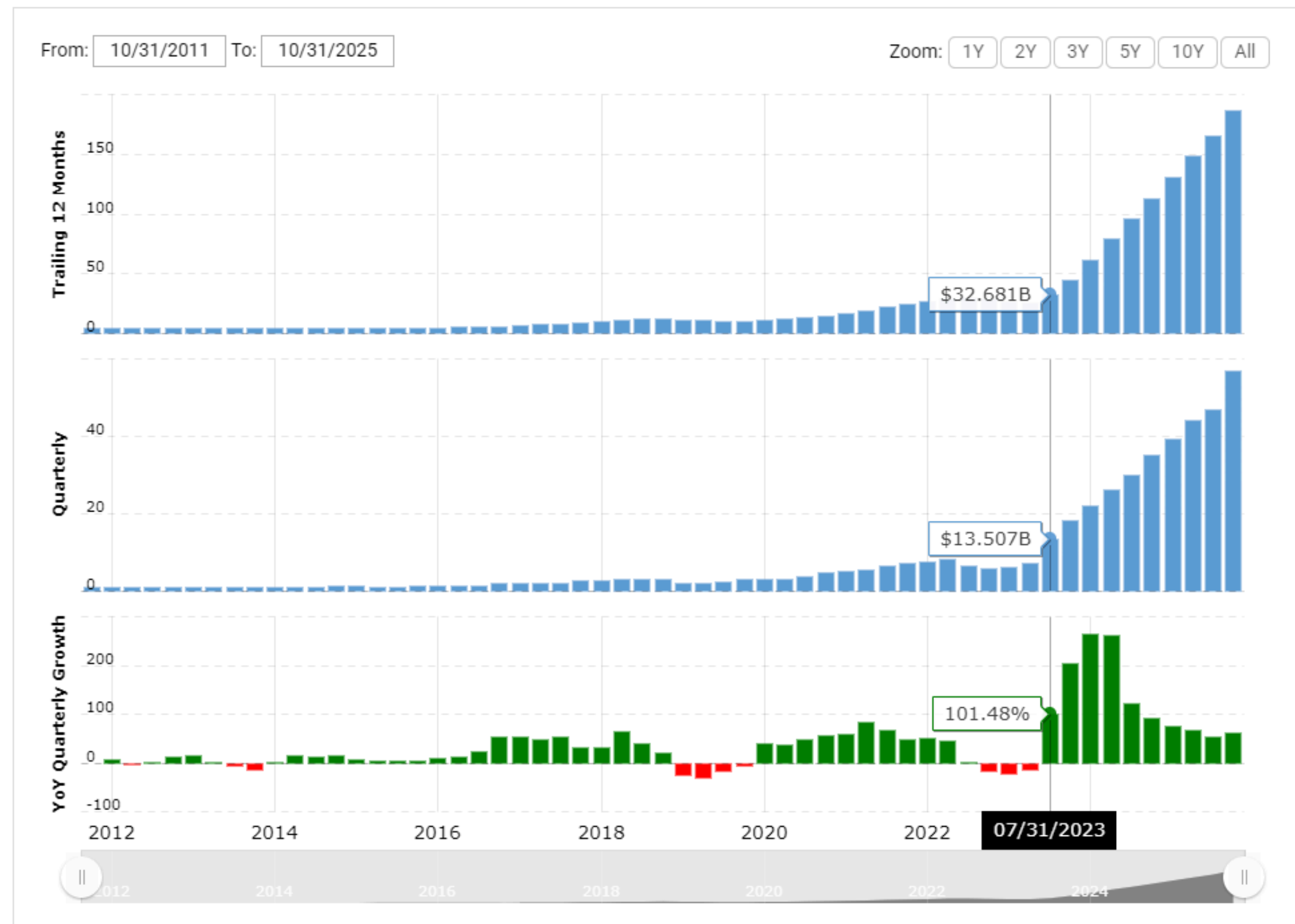


PSY-Monitoring test (Phillips – Shi & Yu, 2020)

Handbook of Statistics, Volume 42 (2020)

It identifies periods of "explosive" growth in asset prices, which is a key characteristic of a financial bubble.

	start	end
1	2015-12-31	2015-12-31
2	2016-05-31	2018-10-31
3	2020-05-31	2022-03-31
4	2023-05-31	2025-11-30



Revenue and profit
Source: macrotrends



Is this growth truly driven by fundamental indicators, or is it partly fueled by external news hype?
Understanding this helps us assess the risks of speculative bubbles



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To determine whether two different NLP-based indicators (Sentiment Analysis, FinBert, and Topic Modeling, BERTopic) predict the movement of NVIDIA's stock price.

Why is it important?

NVIDIA plays a central role in the **global AI** ecosystem, and its stock has shown explosive growth during the recent AI boom. Understanding whether this growth is partly driven by external AI hype—rather than solely by company fundamentals—helps investors, policymakers, and researchers evaluate market sentiment, risks of overvaluation, and early signs of **speculative bubbles**.

DATASETS



+



+



=



Financial News Market Events
Dataset for NLP 2025

S&P 500 with Financial News Headlines
(2008–2024)

Nvidia news data (2022–2025)
headlines & dates

39509
observations



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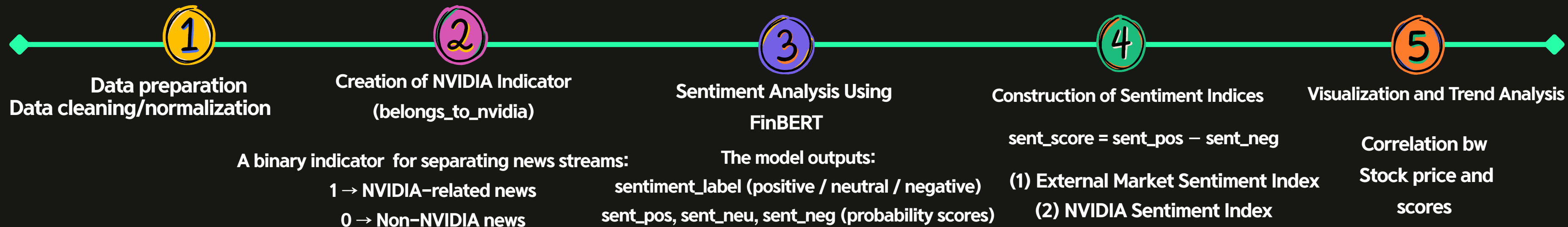


Findings

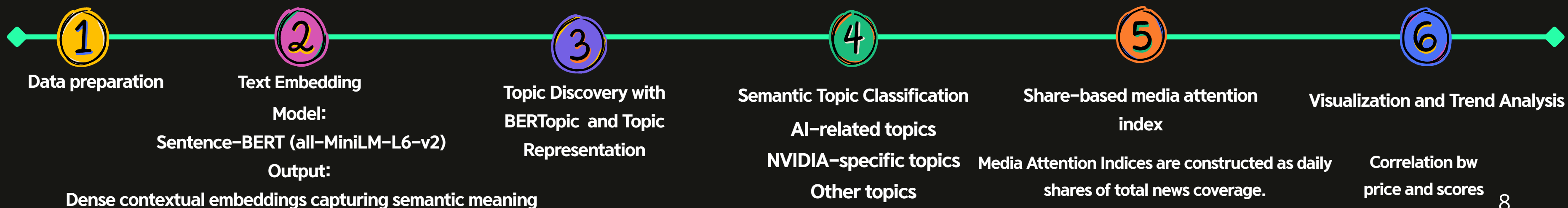


Conclusion

FinBert methods



BERTopic methods



*Each news headline is embedded using a transformer-based semantic model:



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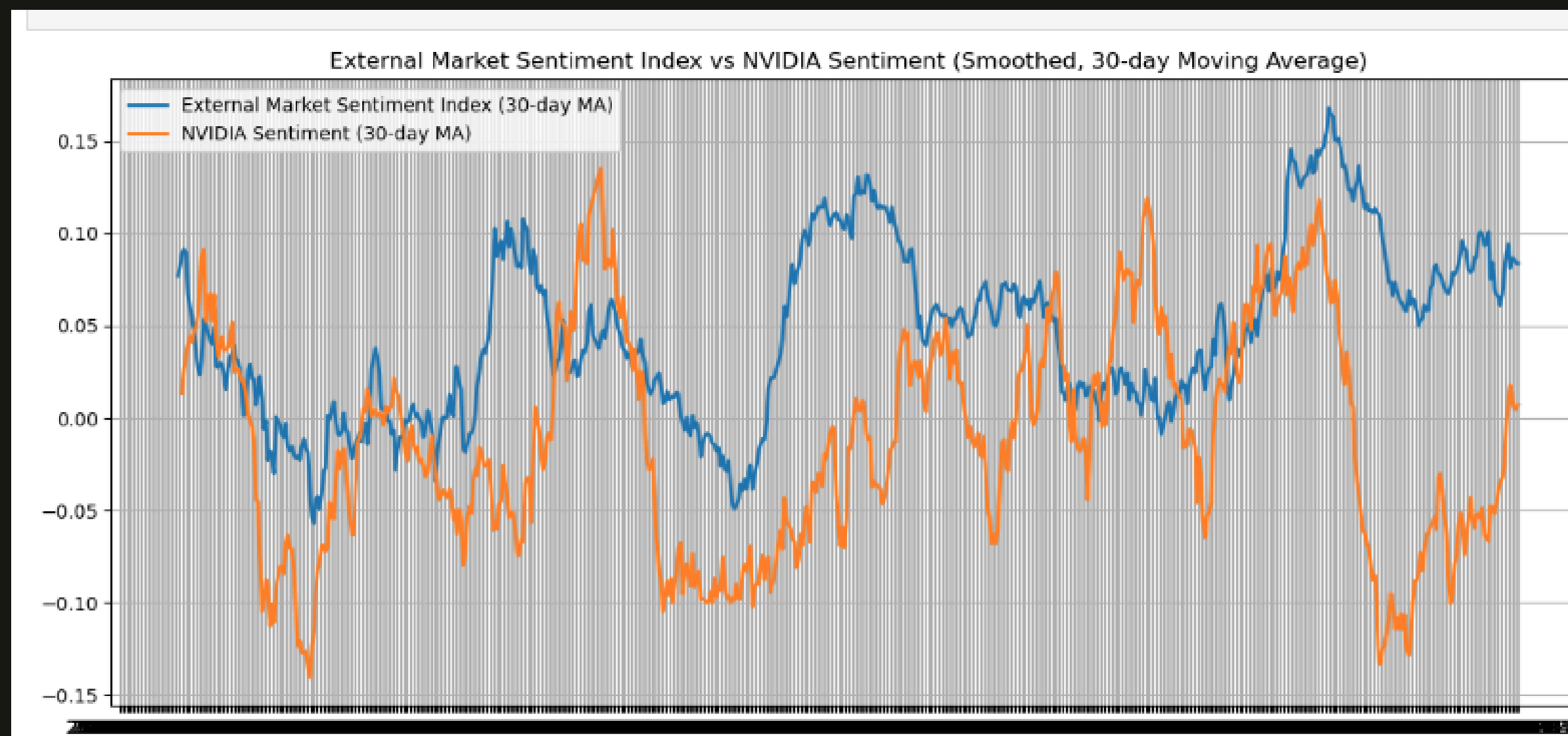


Conclusion

FinBERT results

```
sentiment_label
positive    14005
negative    12525
neutral      8333
Name: count, dtype: int64
```

```
belongs_to_nvidia sentiment_label
0                positive    10956
                  negative     9146
                  neutral     6557
1                negative     3379
                  positive     3049
                  neutral     1776
Name: count, dtype: int64
```



```

=== Correlation with Price ===
      External_Sentiment  NVIDIA_Sentiment  NVDA_Close
External_Sentiment      1.000000      0.162554      0.114582
NVIDIA_Sentiment        0.162554      1.000000      0.236194
NVDA_Close              0.114582      0.236194      1.000000

```

```

=== Correlation with Log Price ===
      External_Sentiment  NVIDIA_Sentiment  log_price
External_Sentiment      1.000000      0.162554      0.100828
NVIDIA_Sentiment        0.162554      1.000000      0.213033
log_price              0.100828      0.213033      1.000000

```

```

=== Correlation with Returns ===
      External_Sentiment  NVIDIA_Sentiment  NVDA_return
External_Sentiment      1.000000      0.162554     -0.059063
NVIDIA_Sentiment        0.162554      1.000000      0.023268
NVDA_return             -0.059063      0.023268      1.000000

```

```

=== Lag Correlation (Sentiment → NVDA_Return) ===
   lag  corr_ext  corr_nvda
0     0 -0.059063  0.023268
1     1  0.005496 -0.010332
2     2 -0.035992  0.028696
3     3  0.001348  0.024223
4     4 -0.012697 -0.003938
5     5  0.044669 -0.011530
6     6 -0.046822  0.046659
7     7 -0.026832  0.009304

```

```

=====
GRANGER TEST: External_Sentiment → NVDA_return
=====
Lag 1: p-value = 0.9844
Lag 2: p-value = 0.7146
Lag 3: p-value = 0.8006
Lag 4: p-value = 0.8629
Lag 5: p-value = 0.7812
Lag 6: p-value = 0.7165
Lag 7: p-value = 0.7712
Conclusion: NO CAUSALITY

```

```

=====
GRANGER TEST: NVIDIA_Sentiment → NVDA_return
=====
Lag 1: p-value = 0.8391
Lag 2: p-value = 0.7923
Lag 3: p-value = 0.8600
Lag 4: p-value = 0.9398
Lag 5: p-value = 0.9701
Lag 6: p-value = 0.8899
Lag 7: p-value = 0.9391
Conclusion: NO CAUSALITY

```

Key findings

Neither External_Sentiment nor NVIDIA_Sentiment is correlated with NVDA returns.

Lag correlations (0–7 days) do not show any structured pattern.

Granger causality tests show no predictive power for both sentiment measures.

→ FinBERT sentiment does NOT predict NVIDIA price movements.

This suggests that sentiment reflects a reaction to past events, rather than being a signal for future price movements.



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BERTopic results

449 topics:

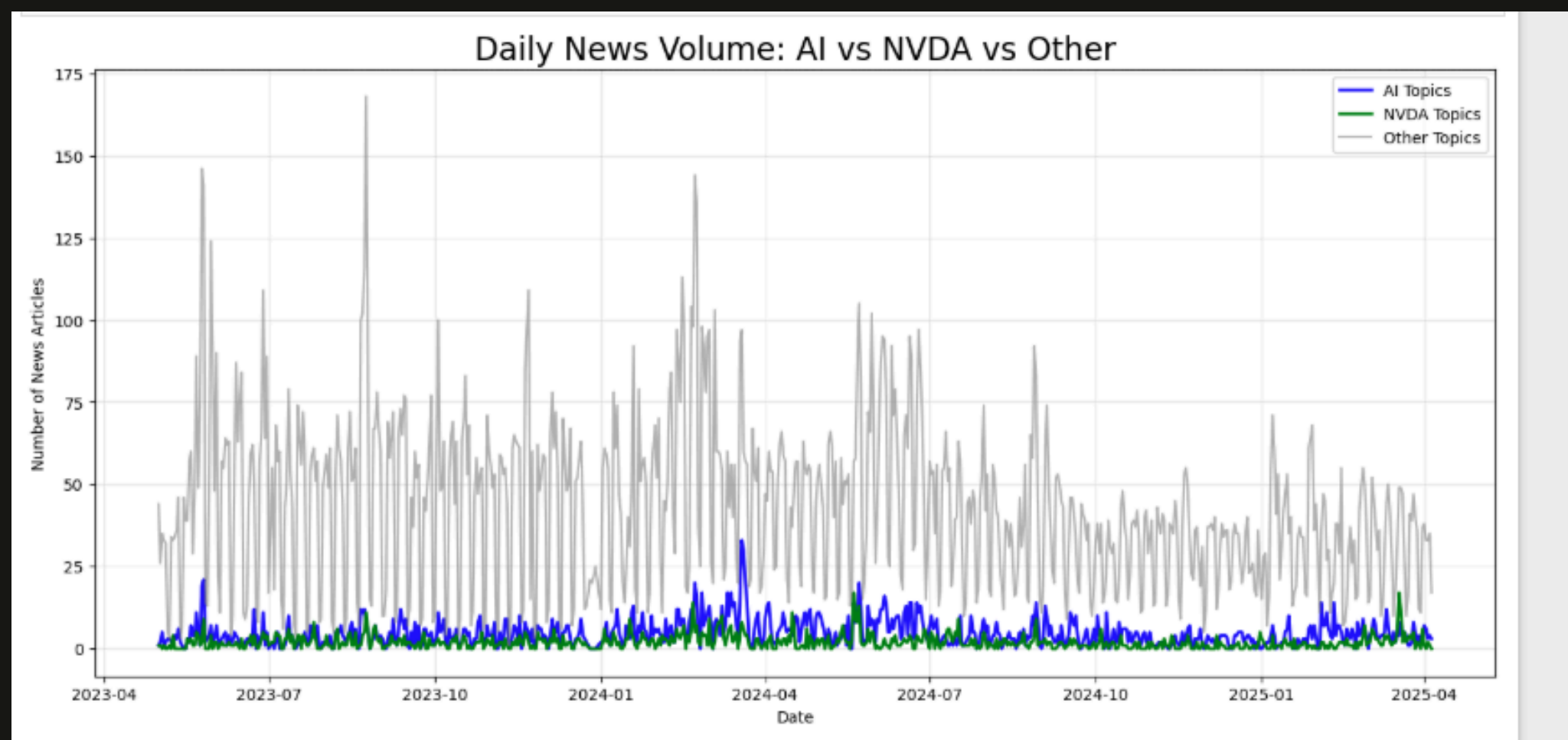
AI topics – 24 topics

Nvidia topics – 12 topics

Others – 413 topics

```
# Keywords for AI-sector topics
ai_keywords = [
    "ai", "artificial", "intelligence", "machine learning", "ml",
    "deep learning", "neural", "llm", "openai", "chatgpt", "deepseek",
    "compute", "computing", "datacenter", "data center",
    "semiconductor", "chip", "chips", "gpus", "gpu demand"
]

# Keywords for NVIDIA-specific topics
nvda_keywords = [
    "nvidia", "nvda", "geforce", "h100", "h200", "cuda",
    "rtx", "gtc", "tensor core", "dgx", "jensen", "huang"
]
```



=== Pearson Correlation with NVDA Price ===			
	AI_share	NVDA_share	NVDA_Close
AI_share	1.000000	0.030289	0.207418
NVDA_share	0.030289	1.000000	0.026506
NVDA_Close	0.207418	0.026506	1.000000

=== Correlation with Log Price ===			
	AI_share	NVDA_share	log_price
AI_share	1.000000	0.030289	0.250011
NVDA_share	0.030289	1.000000	0.059999
log_price	0.250011	0.059999	1.000000

GRANGER TEST: AI_share → NVDA_return	
Lag 1: p-value = 0.8335	
Lag 2: p-value = 0.2966	
Lag 3: p-value = 0.1376	
Lag 4: p-value = 0.2537	
Lag 5: p-value = 0.3376	
Lag 6: p-value = 0.2498	
Lag 7: p-value = 0.3116	
Conclusion: NO CAUSALITY detected	

GRANGER TEST: NVDA_share → NVDA_return	
Lag 1: p-value = 0.0457	→
Lag 2: p-value = 0.0942	
Lag 3: p-value = 0.1906	
Lag 4: p-value = 0.3050	
Lag 5: p-value = 0.2343	
Lag 6: p-value = 0.3685	
Lag 7: p-value = 0.4206	
Conclusion: CAUSES	

AI_topic_share

shows no meaningful correlation with NVDA returns at any lag

does not predict NVDA returns

does not pass the Granger causality test

→ AI hype reflects general news activity but does not drive NVDA price movements.

NVDA_topic_share

exhibits its highest lagged correlation at lag = 1

passes the Granger causality test at lag = 1

→ News attention specifically focused on NVDA predicts a small but statistically significant increase in NVDA returns one day later.

=== Correlation with Returns ===			
	AI_share	NVDA_share	NVDA_return
AI_share	1.000000	0.030289	-0.003276
NVDA_share	0.030289	1.000000	-0.055532
NVDA_return	-0.003276	-0.055532	1.000000

lag	corr_AI	corr_NVDA
0	-0.003276	-0.055532
1	-0.007705	0.079411
2	0.056052	-0.020439
3	0.071919	-0.004892
4	0.007064	0.017197
5	0.032045	-0.051217
6	-0.040708	0.006420
7	0.020037	-0.038714

AI news does not influence NVDA price dynamics — it is too broad and non-specific.

NVDA-focused news has predictive power, suggesting that attention and coverage specifically about the firm drive short-term return movements.

This pattern aligns with behavioral finance literature on attention-induced trading and short-term price pressure.



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Comparison of FinBert and BERTopic

Model	Signal type	Correlation	Granger test	Predictive Power
FinBERT (Nvidia)	sentiment	✗ none	✗ none	no predictive signal
BERTopic (AI topic)	Topic share	✗ none	✗ none	no predictive signal
BERTopic (NVDA topic)	Topic share	✓ weak at lag = 1	✓ yes	weak but real predictive value

→ When the share of NVIDIA-focused news increases, NVDA returns tend to rise the next day.
This is a weak but consistent predictive signal.

Overall, conclusion

1

FinBERT sentiment $\neq$ price prediction

The emotional tone of the news does not drive NVDA's returns.
It reflects reactions to past events, not indications of future price movement.

BERTopic provides weak but meaningful predictive power

2

Topic concentration—especially the share of NVIDIA-related news—captures the arrival of informational shocks.

These shocks influence markets, which is why the NVDA_topic_share has real predictive value.

Why this happens:

- spikes in NVIDIA-topic news correspond to real events (earnings, product releases, partnerships),
- these events affect investor behavior,
- therefore, topic share $\rightarrow$ contains weak forward-looking information.

3

Topic modeling captures market-relevant information better than sentiment analysis.

Future works – LSTM for forecasting the price movements

THANK YOU

APPENDIX

PSY-Monitoring (Phillips – Shi & Yu, 2020)

It identifies periods of "**explosive**" (**exponential**) **growth** in asset prices, which is a key characteristic of a financial bubble.

How It Works:

It analyzes financial data using a "rolling window" approach. In each window, it performs a specialized statistical test to check for growth that is too rapid to be normal. When the test statistic consistently exceeds a certain threshold, it signals the presence of a bubble.

Key Idea:

- It's based on recursive right-tailed ADF regressions.
- Instead of testing only one window, the PSY test:
- Rolls through different time windows of the data.
- Computes ADF statistics for each subsample.
- Looks for periods where the statistic exceeds a simulated critical value → indicating explosive root (bubble).

- **Right-tailed ADF (used in PSY):**

Null: the series has a unit root.

Alternative: the series is **explosive** (root > 1, instead of < 1).

👉 So the PSY test is essentially running *many right-tailed ADF regressions* across rolling windows. It checks if the autoregressive coefficient ϕ in

$$y_t = \phi y_{t-1} + \varepsilon_t$$

Real time monitoring of asset markets:

Bubbles and crises.

is greater than 1 (explosive), rather than just testing if $\phi < 1$.

CORRELATION MEANING

Correlation is a statistical measure that tells us how closely two different things are related and in what direction they move together. It measures the strength and direction of a linear relationship between two variables. The relationship is summarized by a number, the correlation coefficient, which is always between -1 and +1.

Coefficient Value (<i>r</i>)	Type of Correlation	Meaning
Close to +1	Positive (Direct)	The two things move in the same direction . If one goes up, the other tends to go up too.
Close to -1	Negative (Inverse)	The two things move in opposite directions . If one goes up, the other tends to go down.
Close to 0	Zero/No	There is no consistent linear relationship between the two things.

Correlation shows a relationship, but it does not prove that one variable causes the other.

Example: Price and News Hype Trend

Imagine you are looking at the stock price of a company (Variable 1) and the volume of positive news articles or social media mentions (the "hype") about that company (Variable 2).

Positive Correlation: If the stock price tends to rise when the hype volume increases, they have a positive correlation. This suggests they are moving in sync.

Negative Correlation: If the stock price tends to fall when the hype volume increases, they have a negative correlation. This is less common but could happen if the news is viewed skeptically.

Zero Correlation: If the stock price goes up and down with no consistent pattern related to the hype, they have no correlation.

GRANGER TEST MEANING

The Granger Causality Test is a statistical tool used with time series data . It helps determine if past values of one variable are useful for predicting the future values of another variable.

It specifically checks if one time series (Variable X) is a good predictor for another time series (Variable Y). If Variable X's past values significantly improve the prediction of Variable Y's future values (better than just using Variable Y's own past), we say that Variable X "Granger-causes" Variable Y.

Like correlation, "Granger Causality" does not prove a true, physical cause-and-effect relationship.

It only proves a predictive relationship based on timing. The cause must happen before the effect.

EXAMPLE

We want to know: **Does positive news hype predict the stock price changes?**

We would run the test to see if knowing yesterday's Hype value helps us predict today's Price better than if we just knew yesterday's Price alone.

We also can check the reverse: Does knowing yesterday's Price help us predict today's Hype?

The test gives **four possible results:**

- Hype Granger-causes Price: The news hype is useful for forecasting the stock price.
- Price Granger-causes Hype: The stock price changes are useful for forecasting the news hype.
- Both Granger-cause each other: There's a feedback loop.
- Neither Granger-causes the other: They might be correlated, but past values of one don't help predict the future of the other.